Time-Aware Pedestrian Route Planning to Reduce Sun Exposure in Dhaka: A Field Validation Study

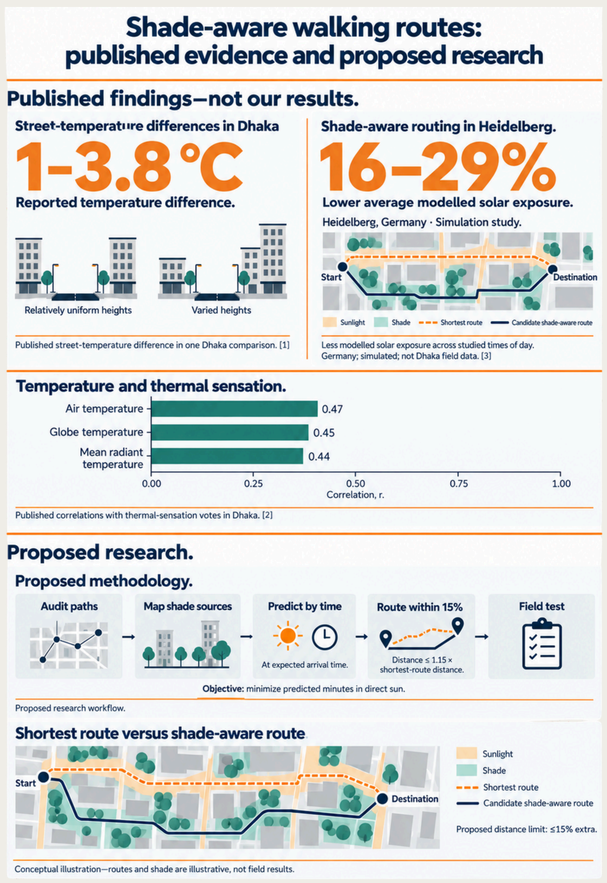
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Introduction & problem

Shade on a walking route changes with the time of day. The shortest path may include sunlit streets even when a slightly longer shaded path is available. Dhaka studies have measured differences in street microclimate, but they do not show whether time-aware routing reduces a walker's observed minutes in direct sun (Sharmin et al., 2015).



Objective

Develop and field-test a time-aware shade route planner for audited streets in two Dhaka neighbourhoods. Compare its routes with shortest-distance and static vegetation routes while limiting detours to 15%.

Research question

Can a route with at most a 15% distance detour reduce observed time in direct sun at a given departure time?

Methodology

Check walking routes
OpenStreetMap and on-site checks
Identify paths and crossings people can actually use.

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Map possible shade
Sentinel-2 vegetation data and observations of trees and buildings
Locate features that may shade each path.

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Calculate the sun's position
Solar-position software
Determine the sun's direction for the date and time of a walk.

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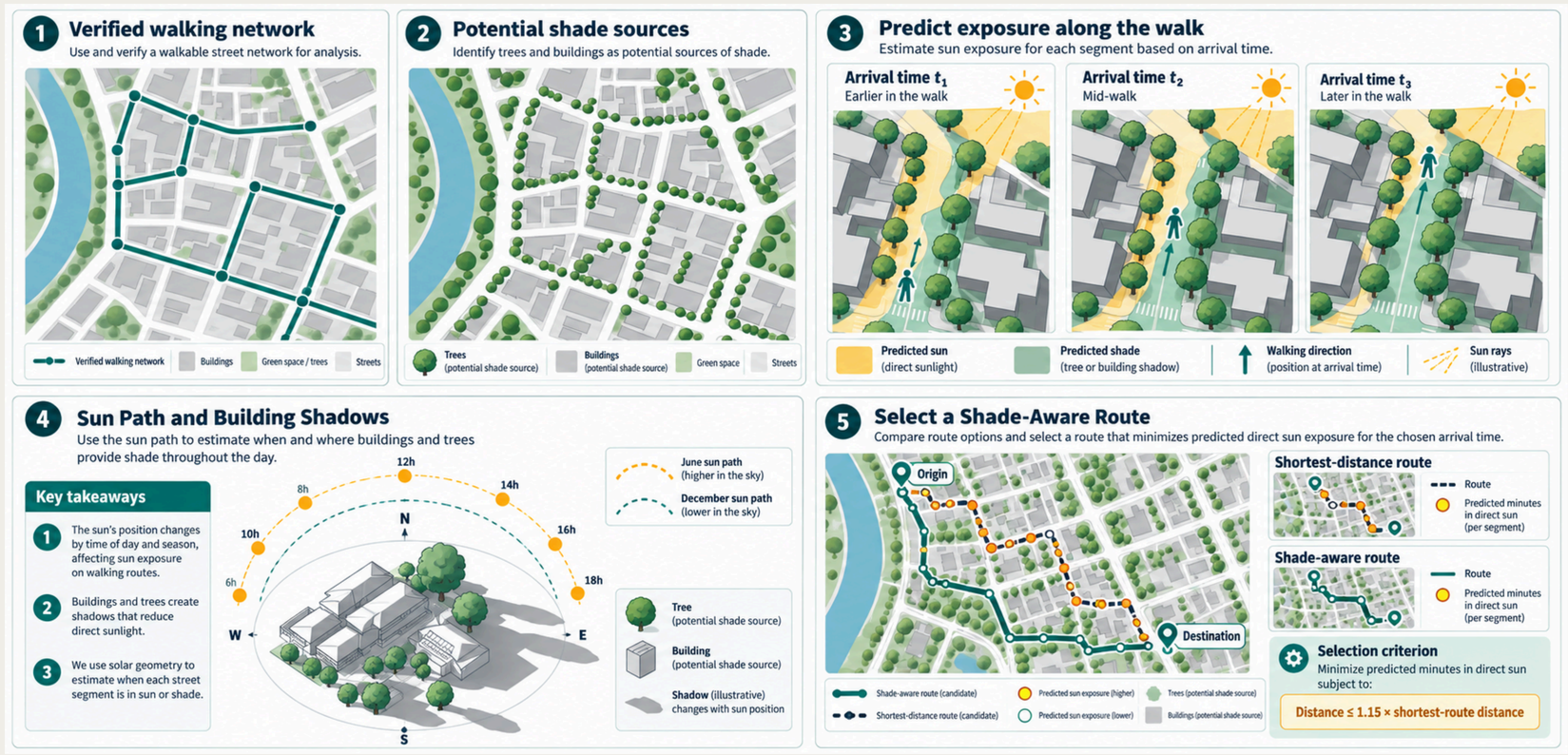
Predict sun or shade
Combine sun position with street direction and observed shade sources
Estimate whether each path segment will be sunlit when the walker reaches it.

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Choose a route
Shortest-path calculation followed by shade-aware route search
Select the route with the fewest predicted minutes in direct sun, allowing no more than 15% extra distance.

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Test the route
Field observations on separate streets and days
Compare predicted shade with actual shade, and compare the recommended route with shortest-distance and vegetation-only routes.



Routing rule: First, find the shortest walkable route. Then, consider routes that are no more than 15% longer. From those routes, choose the one with the fewest predicted minutes in direct sun at the selected departure time.

Expected outcomes and evaluation

Expected outcome	How we will evaluate it
Street-shade dataset and audited walking map	Check recorded sun/shade observations and verify that mapped paths and crossings are usable.
Time-aware shade prediction model	Measure precision, recall, and F1 score on streets and days excluded from model development.
Shade-aware route planner	Compare field-observed **minutes in direct sun**, walking distance, and walking time against shortest-distance and static vegetation routes.
Evidence about benefits and limitations	Report differences between routes, variation across survey periods, and how often a useful alternative exists within the **15% distance limit**.

Fair comparison: Use the same starting point and destination, with comparable departure times and weather conditions.
Expected contribution: Establish whether time-aware routing can reduce observed sun exposure in the surveyed Dhaka streets, and identify the conditions where it succeeds or fails.

Limitations

- Missing or inaccurate walking-path, building-height, and tree data may affect shade predictions.
- Clouds, crossing delays, and temporary obstructions may change actual sun exposure.
- Findings from two neighbourhoods and sunny survey days may not represent all of Dhaka or every season.
- Minutes in direct sun alone cannot establish thermal comfort or health benefits.

Conclusion / intended contribution

This proposed study will test whether time-aware routing can reduce pedestrians' observed minutes in direct sun while allowing no more than 15% extra walking distance. Field validation will determine when the approach helps and where its predictions or route recommendations fail.

References

[1] Sharmin et al. (2015). Building and Environment. https://doi.org/10.1016/j.buildenv.2015.10.007
[2] Sharmin et al. (2019). Energy and Buildings. https://doi.org/10.1016/j.enbuild.2019.05.064
[3] Kolaxidis et al. (2025). Transactions in GIS. https://doi.org/10.1111/tgis.70110